

CANDIDATE GENERATION OR CONSTRAINT VERIFICATION?

A controlled follow-up on LIMIT+ | Independent research brief

On LIMIT+'s literal synthetic grammar, an unmodified BM25 top-1,000 pool has macro candidate recall 0.981 (95.5% micro coverage). Text-only Qwen reranking raises Recall@100 from 0.837 to 0.904; an oracle-decomposed exact verifier reaches 0.927. Candidate generation is not the sole bottleneck.

Why this test

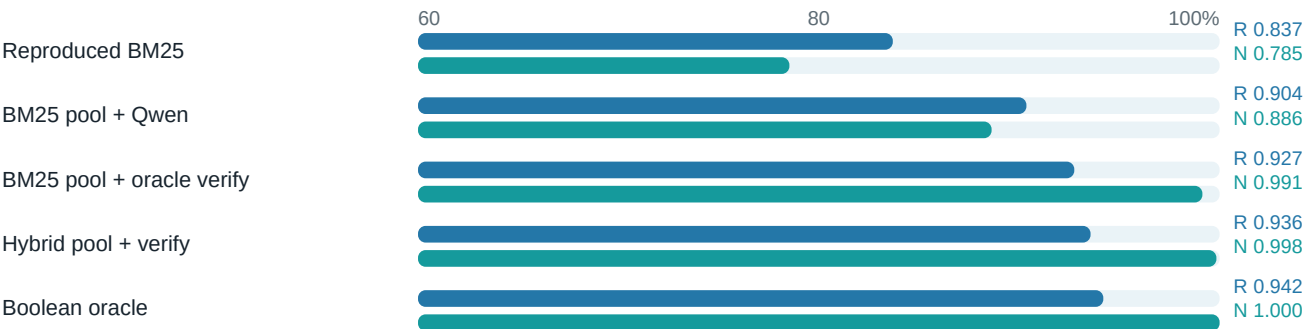
Table 3 evaluates reranking on a gold-complete diagnostic pool containing every relevant document plus sampled BM25 negatives. This follow-up instead keeps an unmodified first-stage BM25 top-1,000 pool and separates **pool coverage** from **constraint verification and ranking**.

Protocol and hard gates

- Pinned full LIMIT (50,000 documents) and released LIMIT+ (700 queries) by revision and SHA-256.
- Independent corpus parser and Boolean executor: exact set equality for 700/700 released qrels.
- Paper-compatible BM25Okapi: case-sensitive NLTK tokens, text only, k1=1.5, b=0.75, epsilon=0.25, _id IDs, released NumPy quicksort ties.
- Every final run is capped at 100 unique IDs; candidate coverage is reported separately; paired template-stratified bootstrap uses 10,000 draws.

Main comparison

Method	R@100	Norm. R@100	nDCG@20	Macro pool R
Paper BM25	0.837	-	0.785	-
Independent BM25	0.837	0.875	0.785	0.981
BM25 pool + Qwen	0.904	0.955	0.886	0.981
BM25 pool + oracle verify	0.927	0.984	0.991	0.981
Hybrid pool + verify	0.936	0.993	0.998	0.989
Boolean oracle	0.942	1.000	1.000	1.000



R = Recall@100; N = nDCG@20. Verification matches BM25 candidate depth, not compute or information access; the hybrid is secondary and adds up to three atomic top-1,000 lists. Raw recall is capped by gold sets larger than 100.

WHAT THE FOLLOW-UP REVEALS

Three findings that refine the bottleneck diagnosis

1. Reproduction is tight.

The released np.argsort(-scores) tie behavior gives $R@100=0.83693$ and $nDCG@20=0.78533$. All six BM25 metrics match the typeset PDF (3 d.p.); at four decimals in the arXiv HTML/source, four match and $R@5/R@100$ differ by $+0.0004/-0.0003$ (ours minus paper).

2. Broad retrieval beats premature composition.

BM25 top-1,000 has macro candidate recall 0.981 (95.5% micro coverage), while hard AND/OR composition of truncated positive atomic lists (with exact NOT filtering) reaches only $R@100=0.639$. Positive-atom-weighted recall is 0.704 at $L=1,000$: large postings cause early pruning.

3. A residual same-pool ranking/verification gap remains.

Across all 700 queries, Qwen adds $+0.067 R@100$ and $+0.101 nDCG@20$ over BM25, yet exact ordering adds another $+0.024$ and $+0.105$. Qwen closes 74% of the recall gain but only 49% of the early-ranking upper-bound gain.

Template view

Template	BM25 N@20	Qwen N@20	Exact N@20	Macro pool R
A	0.922	0.990	0.990	0.990
A OR B	0.934	0.983	0.996	0.995
A OR B OR C	0.951	0.993	0.999	0.998
A AND B	0.992	0.993	1.000	1.000
A AND B AND C	0.994	0.845	1.000	1.000
(A AND B) NOT C	0.175	0.716	0.975	0.905
A NOT B	0.530	0.682	0.980	0.980

Paired uncertainty

Paired comparison	Delta	95% CI	Positive draws
Qwen - BM25 $R@100$	$+0.0667$	$[+0.0561, +0.0774]$	10,000 / 10,000
Qwen - BM25 $nDCG@20$	$+0.1009$	$[+0.0867, +0.1152]$	10,000 / 10,000
Exact - Qwen $R@100$	$+0.0239$	$[+0.0195, +0.0288]$	10,000 / 10,000
Exact - Qwen $nDCG@20$	$+0.1052$	$[+0.0949, +0.1158]$	10,000 / 10,000

Defensible conclusion

On LIMIT+'s synthetic literal grammar, candidate generation is not the sole source of top-100 error. Qwen recovers much of the recall gap, but about half the early-ranking gap remains. Aggregate gains hide a crossover: Qwen helps negation but underperforms BM25 on the 100-query three-way-conjunction slice.

Full neural test

Pinned Qwen3-Reranker-4B scored 700,000 retrieval-derived pairs from query and document text only: no gold injection or truncation. Under the balanced template mix, negation accounts for 98.2% of the signed net macro $nDCG@20$ gain; over 500 non-negation queries the $+0.003$ mean 95% interval crosses zero. Operator and cardinality are confounded. Exact verification is an information-advantaged diagnostic upper bound.

Open question: Would retrieval-pool oracle ordering plus operator-specific neural results help separate candidate ceiling, constraint execution, and model calibration in a follow-up to Table 3?

Sources: arXiv:2605.03824; github.com/informagi/Complex-Set-Compositional-IR; huggingface.co/datasets/orionweller/LIMIT (CC BY 4.0). LIMIT+ has no explicit root license; this brief redistributes no data.